DEV BHODIA

Dubai, United Arab Emirates | +971 55 785 7806 | devbhodia17@gmail.com
linkedin.com/in/devbhodia | devbhodia.com

PROFESSIONAL SUMMARY

Final-year Computer and Autonomous Systems engineering student (BEng, University of Wollongong in Dubai, graduating June 2027) working where wireless, embedded hardware and AI meet. First author of a published IEEE conference paper on deep learning for Wi-Fi channel state information (CSI) sensing; hands-on RF and antenna design experience; and a founder who has taken an AI-powered assistive wearable from concept to field testing with real users. Background spans CubeSat avionics, edge AI and PCB design, built through internships, national competitions and a grant-funded startup. Proficient in Python, MATLAB and ROS 2.

EDUCATION

Bachelor of Engineering (BEng), Computer and Autonomous Systems | University of Wollongong in Dubai (UOWD) | Dubai, UAE | Expected Jun 2027

- Relevant coursework: signals and systems, electromagnetics, embedded systems, digital systems, control systems, programming autonomous systems (ROS 2), image processing for autonomous systems (MATLAB), electronics, circuit analysis, engineering design and management.

Artificial Intelligence and Cybersecurity Track, Summer School | Khalifa University | Abu Dhabi, UAE | May 2026

- Relevant coursework: cybersecurity fundamentals, red teaming and penetration testing, network security, vulnerability assessment, and applied artificial intelligence.

TECHNICAL SKILLS

AI and Data: Machine learning and deep learning (CNNs, attention mechanisms), computer vision (object detection, OCR), edge AI and on-device inference, signal processing and denoising, data analysis, LLM and generative AI APIs

Wireless and RF: Wi-Fi CSI sensing, RF and antenna design, full-wave electromagnetic simulation, return loss, gain and efficiency analysis, LoRa and wireless communication modules

Embedded and IoT: Microcontrollers (MCU), FPGA, Raspberry Pi Compute Module, NVIDIA Jetson, sensor integration (LiDAR, time-of-flight, cameras, EMG), PCB design, fabrication and testing, hardware validation

Software and Tools: Python, MATLAB, ROS 2 (Humble), KiCAD, Fusion 360, AutoCAD, Multisim, Figma, Microsoft Office Suite

EXPERIENCE

Founder and Lead Engineer | Vantage Vision (VANTAGE) | Dubai, UAE | Apr 2025 - Present

- Lead hardware, embedded and AI development of Halo, an assistive wearable that gives blind and visually impaired users spatial awareness through on-device object detection, text recognition and audio-haptic feedback.
- Integrate cameras, LiDAR and multi-zone time-of-flight sensors with a Raspberry Pi Compute Module 5, and build a companion mobile app for device pairing and family location sharing.
- Partner with the Al Noor Centre for Research, Innovation and Development (CRID) and its therapists and users to gather requirements, run user tests and turn feedback into design changes.
- Secured an AED 50,000 development grant from the Expo City Dubai Foundation (Changemakers Academy, selected from 148 applications); coordinate a multidisciplinary team against grant milestones; featured in Khaleej Times and MENA FN.

Embedded Systems Engineering Intern | SpacePoint | Dubai, UAE | Jul 2025 - Jul 2026

- Supported senior engineers on FPGA and MCU integration for the CubeSat Command and Data Handling Subsystem (CDHS) across design, build, integration and test.
- Designed, fabricated and tested PCBs, reviewing schematics and layouts against requirements before build.
- Carried out subsystem testing, hardware validation and troubleshooting, and maintained test records and technical documentation; took part in design reviews on status, issues and interfaces.

Antenna Engineering Intern | University of Wollongong in Dubai (UOWD) | Dubai, UAE | Jun 2025 - Jan 2026

- Designed and simulated antennas for wireless and wearable applications, iterating designs against defined performance targets.
- Used full-wave electromagnetic simulation to evaluate return loss, gain, efficiency and on-body detuning, and presented design iterations in project reviews.

Engineering Peer Tutor | University of Wollongong in Dubai (UOWD) | Dubai, UAE | Oct 2024 - Jan 2026

- Mentored engineering students in core technical subjects and lab work, explaining concepts clearly to students from varied backgrounds.

PUBLICATIONS

D. Bhodia, A. Elkhouly and M. F. A. Malek, "Attention-enhanced CNN with wavelet denoising for through-wall human activity recognition using Wi-Fi CSI," *International Conference on Electrical and Electronics Engineering (ICEEE)*, 2025.

ENGINEERING PROJECTS

Through-Wall Human Activity Recognition with Wi-Fi CSI | UOWD Research | Published at ICEEE 2025

- Used Wi-Fi channel state information as a device-free sensor to recognise human activity through walls, applying wavelet denoising and an attention-enhanced CNN classifier.

Kilmara - Edge AI for HVAC Prediction | SREO Program, MBR Centre for Government Innovation | Top 5

- Built an on-device edge AI system that predicts HVAC operating behaviour from a five-parameter sensor model, running real-time inference without cloud dependence to improve building energy efficiency.

ExoHand - EMG Assistive Hand and Mixed-Reality Rehabilitation | UOWD | Emerging Project of Excellence Award

- Developed an EMG-controlled assistive hand for stroke and paralysis rehabilitation, processing dry-electrode muscle signals to drive actuator control, paired with a gamified mixed-reality therapy environment on Meta VR.

LEADERSHIP AND ACTIVITIES

President | UOWD Sustainability Society | Dubai, UAE | Jun 2025 - Present

- Lead a student society running sustainability, coexistence and community initiatives; manage a student committee and liaise with university staff and external partners.
- Society-led initiatives contributed to UOWD receiving the Golden Shield, the highest national tier of the UAE Ministry of Tolerance and Coexistence Tolerance Shield contest (Oct 2025).

Volunteer | Al Noor Association for People of Determination | Dubai, UAE | Aug 2025 - Present

- Volunteer at the annual Family Fair and centre programmes, working directly with people of determination, their families and staff.

Student Intern and Coordinator | Nikon Youth and Kids School | Dubai, UAE | Aug 2025 - Dec 2025

- Coordinated the MOCCAE x Nikon field trip to the Kalba mangrove conservation site end to end, covering planning, logistics, transport, scheduling and on-site delivery.

AWARDS AND RECOGNITION

- Red Bull Basement - UAE National Winner; represented the UAE at the World Final in San Francisco.
- Mohammed bin Rashid Innovation Trailblazers - 3rd Place, AED 15,000 prize (Jan 2026).
- SREO Program, 5th Edition, Mohammed Bin Rashid Centre for Government Innovation - Top 5 (Aug 2026).
- Expo Changemakers Academy, Expo City Dubai Foundation - selected from 148 applications; AED 50,000 development grant.
- Golden Shield, Tolerance Shield Contest - UAE Ministry of Tolerance and Coexistence; awarded to UOWD as the top-performing Tolerance Club, contributed as Society President (Oct 2025).
- Emerging Project of Excellence - Engineering Design and Management 3, UOWD (Feb 2026).
- School of Engineering Innovation Fair 2025, UOWD - Runner-Up Award and two UOWD Platinum Awards (Capstone Project).
- Certificate of Appreciation, Tolerance Clubs Initiative - UAE Ministry of Tolerance and Coexistence (2023-2024).

CERTIFICATIONS AND TRAINING

- Samsung Innovation Campus - Artificial Intelligence Course 2025, Certificate of Completion.
- Innovation Trailblazers - Business Model and Pitching Workshop, Certificate of Completion (Jan 2026).
- Global Cities Summit Young Professionals' Forum - Expo City Dubai and Dubai Youth Council (Oct 2025).